

offline, once per technology (no stack, no netlist involved)

physics results: reference FDM on every rail (truth); SKN + JCX: the fast, differentiable surrogate

kernel data (reference FDM)

random segment profiles $(\theta, r, \rho_j, \lambda)$
96 τ points per profile

SKN training

4×128 SiLU + Fourier features
 $\sqrt{\tau}$ -scaled targets, teacher-densified labels

frozen segment kernels

$s_G, s_M, a(\xi; \theta, r, \rho_j, \lambda; \tau)$
reused by every stack case

per stack case (seconds): stack-level EM sign-off

stack case

dies, block powers,
feeds, package,
rule-sized straps

3-D thermal field

HotSpot 7 3-D grid
+ Joule fin (Γ, T_m)
stack and standalone

grid currents

DC power-grid solve
 j on every rail
segments between feeds

Blech screen

Blech immortality
 $jL < (jL)_c$
feeds and rails

JCX exact closure

Duhamel superposition of junction
fluxes; tridiagonal per time step;
tabulated kernels; batched; autograd

kernel look-up at the
segment end points

outputs: $\sigma(t)$ at every node, t_{nuc} of every rail → the sign-off metric ladder

stack-level sign-off

mortal set, earliest t_{nuc} , stress margin,
Top-k ranking, cross-die alignment

targeted repair

widen only the violating rails
(7.9-22× less metal than blanket)

sensitivities & budgeting

$\partial \log t_{nuc} / \partial P_{block}$ by autograd,
projected-gradient power re-allocation